

Kevin Benny Mathew

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EDUCATION

2023 - 2027	B.E. (Hons.) Electronics and Communication Engineering - BITS Pilani, Goa Campus	8.21/10 (Upto Semester 5)
May 2023	ISC (Class XII) - The Bishop's School, Camp	97.6%
May 2021	ICSE (Class X) - St. Mary's School, Pune	99.0%

WORK EXPERIENCE

Summer Intern

May 2025 - Aug 2025

- Worked at the **Indian Meteorological Department** on sensor fusion and algorithm development.
- Implemented custom fuzzy logic and Kalman filters for high-accuracy temperature sensing.
- Developed embedded code for ESP8266 hardware using C++ and PlatformIO.
- Conducted data analysis using Python and documented results for research publication.

PROJECTS

Development of Flexible Tactile Sensors

Dec 2025 - Present

- Currently in process of developing tactile sensors for robotic perception under Dr. Tushar Sakorikar.
- Project has finished literature review and we are in the process of prototyping a piezo-resistive tactile sensor.

Low Cost Electromechanical Tensile Tester

May 2025 - Dec 2025

- Developed a horizontal tensile tester for the characterization of soft materials under Dr. Tushar Sakorikar.
- Executed end-to-end development including mechanical design, 3D modeling, and electronics architecture.
- Integrated motion control using Teensy microcontrollers, stepper motors, and a custom UI.

Project Vulcan: Humanoid Robot

Feb 2024 - Present

- Led the bionic arm subsystem, developing a 7-DOF biomimetic robotic arm emulating human movement.
- Improved internal mechanical systems for facial movements, specifically eyes and mouth.
- Implemented amplifiers for microphones located in the robot's ears.

Project Kratos: Mars Rover

Aug 2024 - Present

- Core member of the electronics subsystem; migrated the code stack from ROS Noetic to ROS2 Humble.
- Redesigned the electronics box for modularity, easier assembly, and improved debugging.
- Designed custom mounts for cameras and sensors using Fusion 360.

LEADERSHIP & VOLUNTEERING

Research Head - Electronics and Robotics Club (ERC), BITS Goa Apr 2025 - Present

- Guiding **75+** students across multiple robotics projects (Vulcan, Snakebot, Quadruped, Automated Foosball, Automated Air Hockey)
- Managing project progress and integrating electronics, mechanical, and automation verticals.
- Organized and designed ROS2-based hackathon with **200+ participants nationwide**

Committee Member - Sandbox Innovation Lab, BITS Goa Feb 2024 - Present

Maintaining lab equipment of the campus student-run (3D printers, laser cutters) and managing tech club purchases.

Volunteer - Abhigyaan, BITS Goa Oct 2023 - Present

Provided evening tuition classes for K-12 children of campus non-teaching staff.

SKILLS & PROFICIENCY

Languages	C++, C, Python, Java, Verilog
Technologies	ROS/ROS2, Git, Linux, CV, Microcontrollers (ESP32, Arduino, Teensy, ESP8266)
Tools	KiCAD, Fusion 360, Solidworks, Onshape, MATLAB, PlatformIO, Android Studio
Key Courses	Control Systems, Microprocessors & Interfacing, Digital Signal Processing